

Experiment - 10

Name: K. SRI LASYA

Regno: 192472216

Course Code: MLA0305

Slot: B

Aim

To evaluate and compare the performance of different DQN variants, namely **Double DQN (DDQN)**, **Dueling DQN**, and **Prioritized Experience Replay**, using the CartPole reinforcement learning environment.

Algorithm / Procedure

1. Initialize the CartPole environment.
2. Create the neural network for DQN-based learning.
3. Implement Double DQN by selecting actions using the current Q-network.
4. Implement Dueling DQN by separating the value and advantage streams.
5. Implement Prioritized Experience Replay by giving higher priority to experiences with larger TD errors.
6. Train each DQN variant for multiple episodes.
7. Calculate the reward obtained in each episode.
8. Calculate the average reward for each algorithm.
9. Compare the performance of DDQN, Dueling DQN, and Prioritized Experience Replay using graphs.
10. Identify the DQN variant with better performance.

Code

```
!pip -q install gymnasium tensorflow matplotlib
import gymnasium as gym
import tensorflow as tf
import numpy as np
import random
import matplotlib.pyplot as plt
%matplotlib inline
# -----
# 1. Create Environment
# -----
```

```
env = gym.make("CartPole-v1")

state_size = env.observation_space.shape[0]
action_size = env.action_space.n

print("State Size :", state_size)
print("Action Size:", action_size)
plt.plot(
    per_rewards,
    label="Prioritized Experience Replay"
)

plt.xlabel("Episode")

plt.ylabel("Reward")

plt.title(
    "Performance Evaluation of DQN Variants"
)
plt.legend()

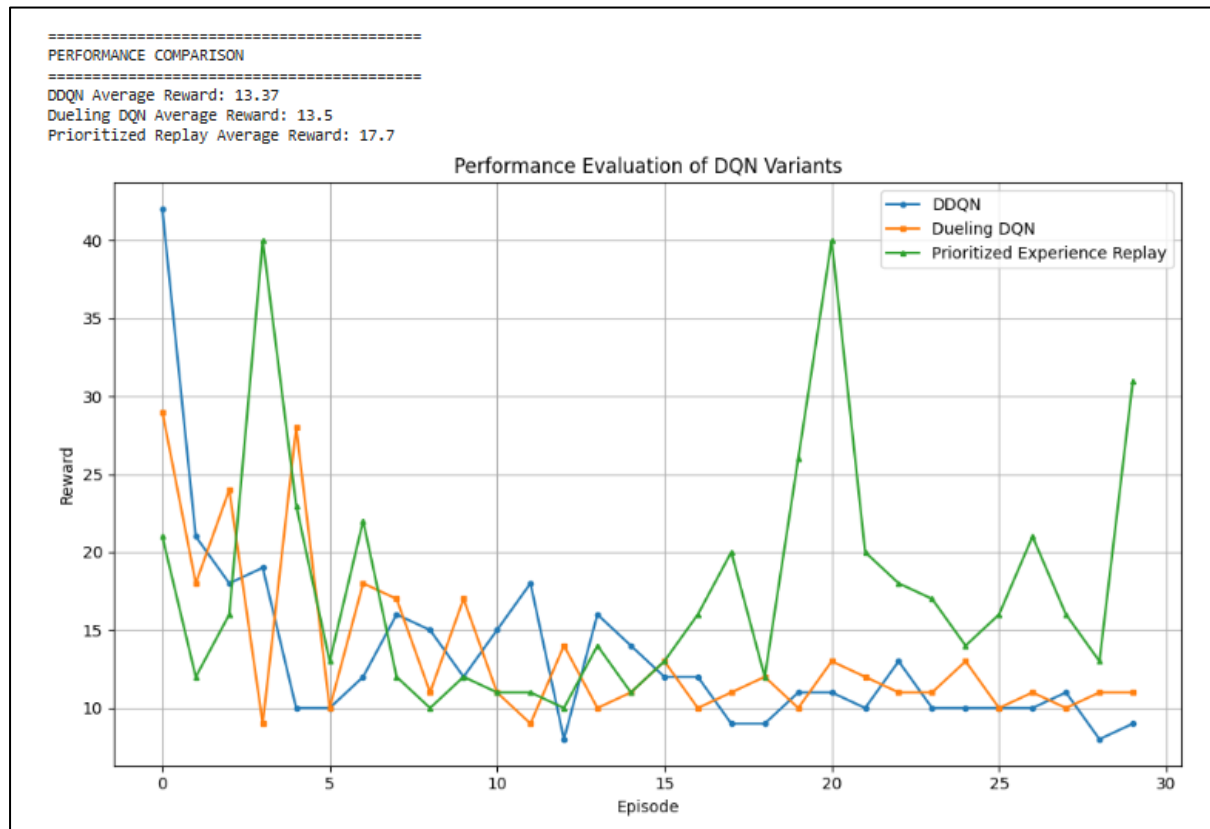
plt.grid(True)

plt.tight_layout()

plt.show()
names = [
    "DDQN",
    "Dueling DQN",
    "Prioritized Replay"
]
```

```
average_rewards = [  
    ddqn_average,  
    dueling_average,  
    per_average  
]  
  
plt.figure(  
    figsize=(8, 5)  
)  
  
plt.bar(  
    names,  
    average_rewards  
)  
plt.xlabel("DQN Variant")  
plt.ylabel("Average Reward")  
plt.title(  
    "Average Reward Comparison"  
)  
plt.grid(  
    axis="y"  
)  
plt.tight_layout()  
plt.show()  
env.close()  
print(  
    "\nExperiment 10 Completed Successfully!"  
)
```

Output



Result

The performance of **DDQN**, **Dueling DQN**, and **Prioritized Experience Replay** was successfully evaluated, and their rewards were compared using graphical analysis.